# Music Sound Interpreter

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# The Rundown

- Who is the client?
- What is the project?
- Why is this being done?
- How will this project be done?
- How much time will it take?
- How much is this project going to cost?

# The Client

- My client is Rosemary
  - Has Down Syndrome
  - Is deaf
  - Communicates by sign language
  - Spends some of her time at Bridgewell in Lynn

# Some Facts about Down Syndrome

- Down Syndrome is a genetic disorder involving the 21st chromosome
- About 1 in 700 people are born with DS
- 400,000 diagnosed people in the U.S.
- Most prominent type is Trisomy 21
  - Accounts for 95% of all cases
  - Cells have an extra copy of chromosome 21

# Effects of Down Syndrome

- Effects vary from person to person
- Possible for diagnosed people to lead normal lives
- Common effects include:
  - Physical disfigurement
  - Slow mental development
  - Increased risk of health issues

# The Problem

- Bridgewell hosts learning activities such as concerts
- Since Rosemary is deaf, she feels the speakers to understand what's going on
- In doing so, she is physically left out of her groups

# The Solution: My Project

- Lap pad that senses audio signals then activates visual effects on one side and vibrational effects on the other
- Contains recorded responses such as: ”That's awesome!”
- Using the Music Sound Interpreter, she will able to join her groups extensively

# Functions

- Visual and vibrational effects controlled by frequency and strength of audio signals
- Visual components consist of a rectangular LED grid
  - Each column represents a different frequency range
  - As intensity changes, LEDs fade in and out
- Vibrational effects activated similarly

# Bonus Purposes

- Can be used to teach individuals how to control their voices
  - Instructor would say something to be copied,
    then the user would then try to match the instructor
- Could also be used as decorative piece

# Designing for Everyone

- Rosemary will most likely not be the only user
- Should be designed so it's usable by anyone
- Design factors include:
  - Safety
  - Accessibility
  - Simplicity

# Physical Specifications

- About as big as an open notebook (12”x21”x1”)
- Weight = 5 lbs
  - The lighter, the better
- Cushioned
- Shatterproof
- Waterproof
- Easily cleanable
- Cased in flexible material

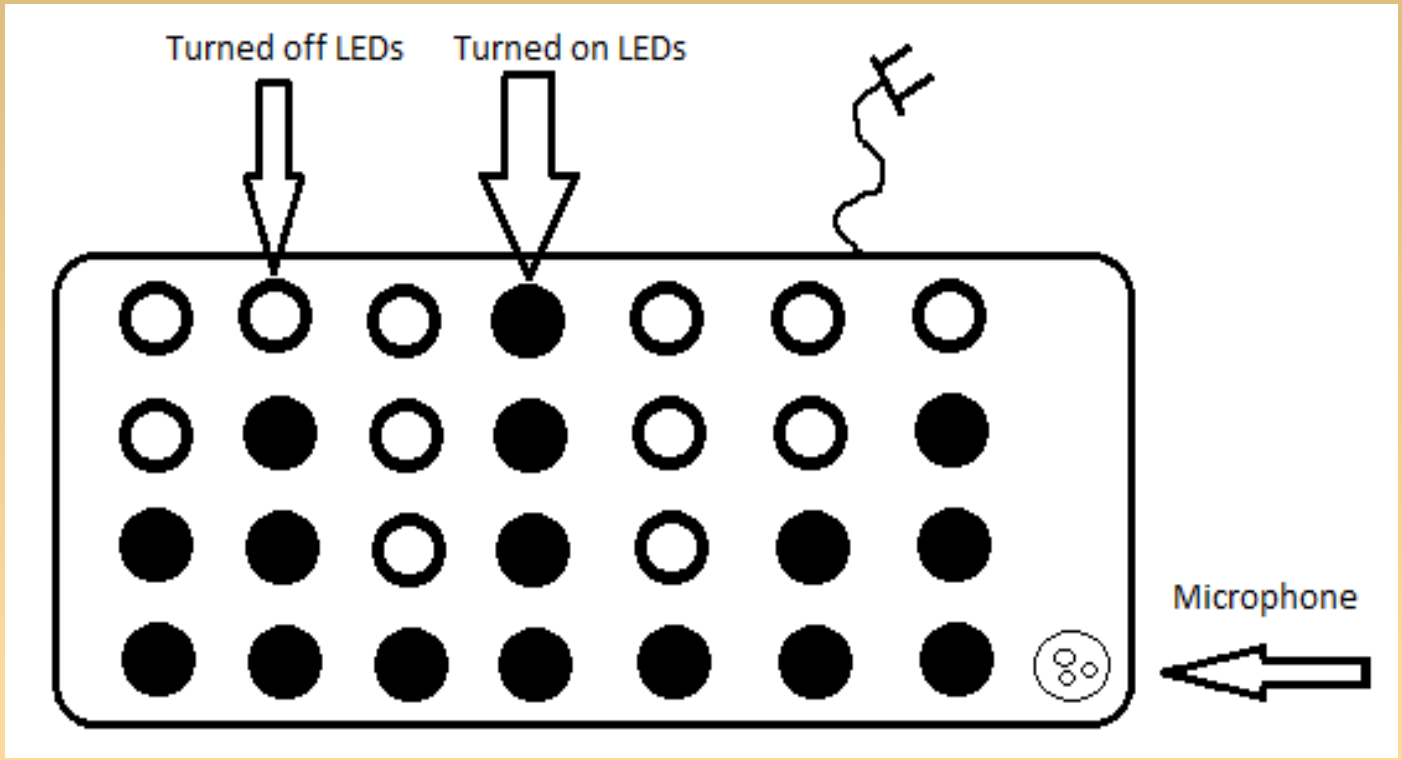
# Electrical Specifications

- Powered via wall outlet
- Includes a function toggle switch
  - Only vibrations, only lights, both on, both off
- Automatic shut-off feature
- Low voltage, about 12V maximum
- Medium current, about 2A maximum

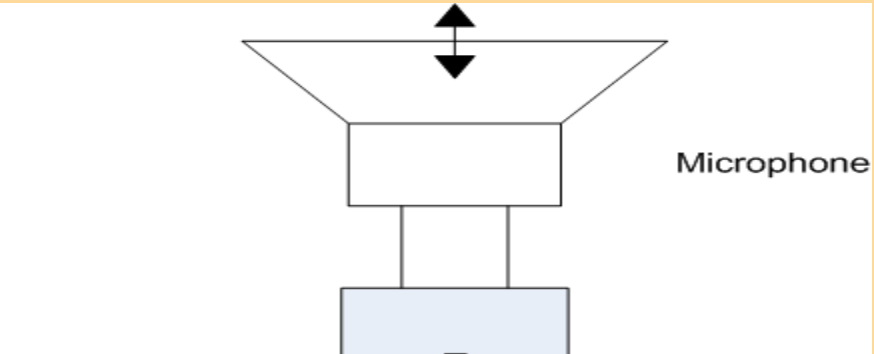
# Effect Specs

- Uses calming colors such as pink, purple and green
  - Avoid using red, yellow, and orange
- Uses smooth visual components
  - No fast-blinking lights
- Intensity of lights and vibrations shouldn't be strong

# Functional Diagram



# Block Diagram

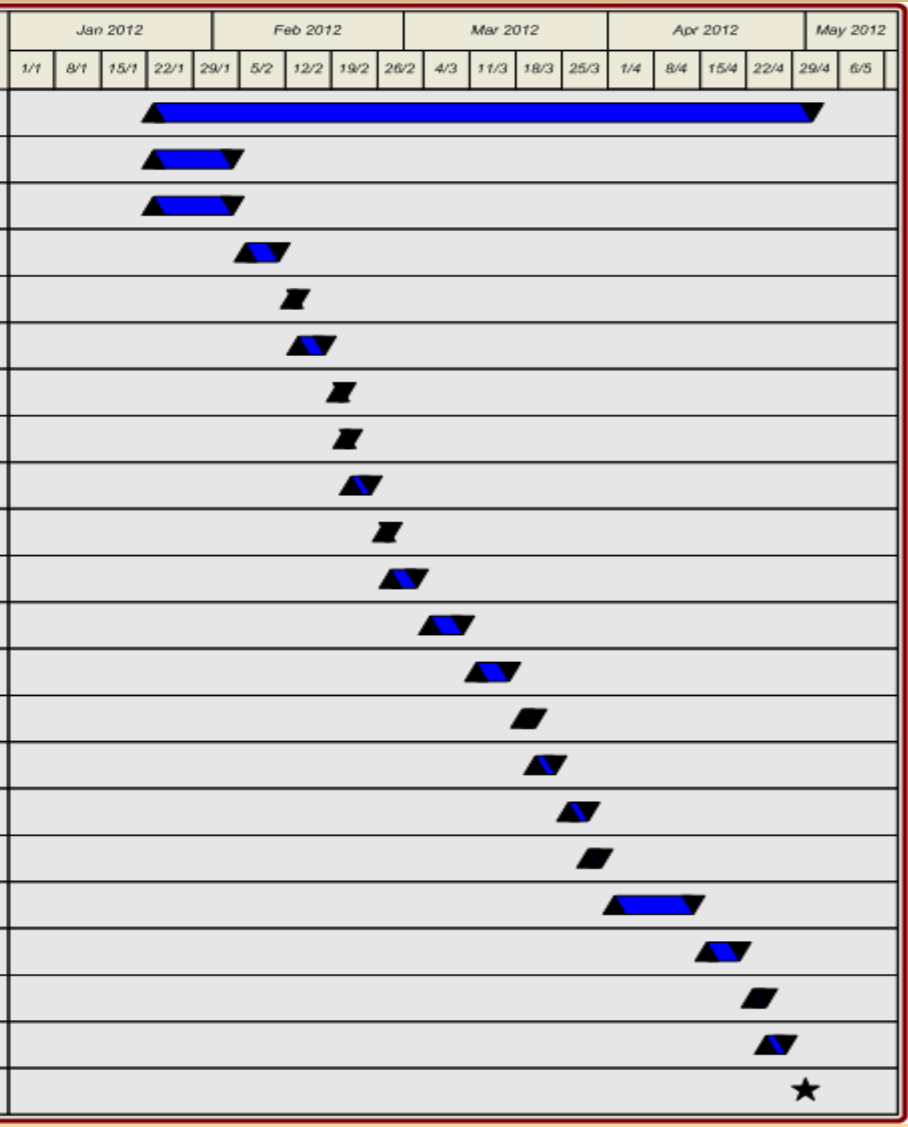


# Creation Procedure

- Design parts from beginning of product activation towards output
- Test parts in descending order
  - First tests are stand alone
  - Second are with previous parts included

# Gantt Chart

| ID | Task Name | Start | Finish | Duration | Hours Worked |
|---|---|---|---|---|---|
| 1 | Total Completion Time | 1/23/2012 | 5/1/2012 | 100d | 193h |
| 2 | Part Hunting | 1/23/2012 | 2/3/2012 | 12d | 20h |
| 3 | Design General Final Circuit | 1/23/2012 | 2/3/2012 | 12d | 20h |
| 4 | Pre-Amp Design | 2/6/2012 | 2/10/2012 | 5d | 10h |
| 5 | Test Pre-Amp connected to Mic | 2/13/2012 | 2/13/2012 | 1d | 4h |
| 6 | Filter Design | 2/14/2012 | 2/17/2012 | 4d | 8h |
| 7 | Stand Alone Filter Test | 2/20/2012 | 2/20/2012 | 1d | 4h |
| 8 | Filter test with Pre-amp and Mic | 2/21/2012 | 2/21/2012 | 1d | 5h |
| 9 | LED Controller Design | 2/22/2012 | 2/24/2012 | 3d | 16h |
| 10 | Stand Alone LED Controller Test | 2/27/2012 | 2/27/2012 | 1d | 4h |
| 11 | Motor Controller Design | 2/28/2012 | 3/2/2012 | 4d | 8h |
| 12 | Stand Alone Motor Controller Test | 3/5/2012 | 3/9/2012 | 5d | 10h |
| 13 | Main Circuit Test | 3/12/2012 | 3/16/2012 | 5d | 10h |
| 14 | Design and Test Voice Module | 3/19/2012 | 3/20/2012 | 2d | 10h |
| 15 | Design Automatic Shut-Off Feature | 3/21/2012 | 3/23/2012 | 3d | 12h |
| 16 | Stand Alone Shut-Off Test | 3/26/2012 | 3/28/2012 | 3d | 6h |
| 17 | Shut-Off Test with Main Circuit | 3/29/2012 | 3/30/2012 | 2d | 4h |
| 18 | Design Supply Circuit | 4/2/2012 | 4/13/2012 | 12d | 20h |
| 19 | Test Final Circuit | 4/16/2012 | 4/20/2012 | 5d | 10h |
| 20 | Assemble Final Product | 4/23/2012 | 4/24/2012 | 2d | 4h |
| 21 | Test Final Product | 4/25/2012 | 4/27/2012 | 3d | 6h |
| 22 | Present Music Sound Interpreter | 5/1/2012 | 5/1/2012 | 0d | |



# Time Analysis

- Beginning Date: January 23, 2012
- Ending Date: May 1, 2012
- Duration: 100 days
- Time spent on project: 193 hours
- Designing each block mostly takes up a week
- Testing usually takes a few days
- Work between 2-4 hours per day

# Market Research

- Most products are very expensive
- Not many found that have similar functions
- Most products have simple, “discrete” functions
- Very few are sound controlled
- Potential for large market, even outside of assistive technology

# Main Competing Product

**Interactive Fanlight**

- Displays lights in a fan-like manner
- Automatic and sound controlled modes
- Multiple display styles within each mode
- Only amplitude dependent
- Priced at $2,675.00

# Labor and Overhead

- Average time worked per week: 13.5 hours
- 14 week long project
- Labor Cost = $5,724
- Overhead Cost= $2,919

# Parts and Prices

# Total Cost

- Direct Labor: $5,724
- Overhead: $2,919
- Parts: $325
- **Total Cost of Project: $8,962**

# Final Words

Any Questions?

Thanks everyone for your time!